Michael Diaz-Stewart

Vancouver, BC | Relocating to Toronto | diaz-@live.ca
Portfolio Website | GitHub | Kaggle | LinkedIn

Applied ML scientist with 3+ years of computer vision research. Builds high-quality, containerized, testable and reproducible PyTorch workflows at scale. Published medical deep learning research and from-zero professional web and mobile applications deployments. Experienced with AI tooling, both for fast iteration or maintaining a codebase.

EXPERIENCE

Graduate Research Assistant | AI in Medicine Lab, UBC Jul 2023 – Oct 2026

- Built PyTorch pipelines for classifying biomarkers from histology, leveraging terabyte scale data and pathology **foundation models** (UNI, Optimus, GigaPath, Phikon). Used **multiple-instance learning** and ensembling
- Evaluated reference stability, **domain shift**, and **interpretability** with multi-institutional healthcare cohorts (TCGA, BC Cancer, MD Anderson)
- Developed a microsatellite instability pre-screening classifier with **94% sensitivity** and **98% negative predictive value** in retrospective external validation
- Domain adapted tumor pseudo-labeling with **supervised finetuning** to improve performance on biopsy samples
- Interpreted models with attention maps and **vision-language models** (PathGen-LLaVA, MedGemma**)**
- Created reproducible, containerized GPU workflows using **Singularity** and **SLURM** on remote access Linux

Clinical ML Research Intern | St. Paul's Hospital Jan 2024 – Aug 2024

- Worked on CT-based **AI screening** for evaluation of pacemaker extraction risk, presented poster research

Senior Software Developer | Stafits Nov 2017 – Oct 2018

- Technical **ownership** of zero-to-one MERN stack website and customer-management backend deployment
- Also built a browser extension and mobile application | Fast paced **startup** environment | Google Places API

PROJECTS (see portfolio for more)

Representation modeling League of Legends

- Developed a **self-supervised** model for game data analysis, comparing **masked reconstruction** learning policies and **transformer**-based architectures. Improves multi-task performance versus supervised transformer baselines.

EDUCATION

Master of Applied Science, Biomedical Engineering | University of British Columbia Defended Oct 2026

Grade: 93.2% | Graduate ML Coursework | NSERC and CIHR scholarships awardee (Over $30,000)

Honours Bachelor of Science, Biology | York University May 2022

First Class Standing; Dean's Honour Roll

PUBLICATIONS & THESIS

1. **Diaz-Stewart M**. Task-Dependent Utility of Computational Pathology in Colorectal Cancer. Master's thesis. University of British Columbia; 2026. 10.14288/1.0457100
2. Hadjifaradji A, **Diaz-Stewart M**, Chu J, et al. A Deep Learning Framework for Classification of Neuroendocrine Neoplasm Whole Slide Images. Cancers. 2025;17(18):2991. 10.3390/cancers17182991
3. Paul A, Mendis S, **Diaz-Stewart M**, et al. Comparison of Lanreotide and Octreotide LAR Use and Outcomes for Gastrointestinal Neuroendocrine Tumors in British Columbia, Canada. Cancer Control. 2026. 10.1177/10732748261417423